

MATERIAL SAFETY DATA SHEET

1. CHEMICAL PRODUCT AND COMPANY IDENTIFICATION

MATERIAL IDENTITY

PRODUCT CODE AND NAME

PROPYLENE GLYCOL

Chemical Name and/or family or description:

Mono Propylene Glycol

COMPANY INFORMATION

Manali Petrochemicals Limited
Ponneri High Road,
Manali
Chennai 600 068
Ph. No. +91442594 1025 /+914425941249

TELEPHONE NUMBERS

Transportation Emergency : +9144-25941550, +9144-25941402

2. COMPOSITION AND INFORMATION ON INGREDIENTS

Composition:

Chemical Name	CAS Number	Range In %
1,2-propanediol	57-55-6	100.00

THIS PRODUCT IS CONSIDERED AS NON-HAZARDOUS.

3. HAZARD IDENTIFICATION

Emergency Overview

Appearance: Colourless liquid

Odour : Odourless

HEALTH HAZARD DATA

Routes of entry	Inhalation; Eye contact; Skin absorption; ingestion.
Effects of exposure / symptoms	Massive ingestion- symptoms similar to alcohol intoxication, inhalation of high vapour conc. May cause CNS depression with headaches, dizziness & drowsiness.
Emergency treatment	Remove from exposure & remove contaminated clothing, if unable to breath, give Oxygen. Incase of accidental ingestion, give plenty of water to drink, then induce vomiting & obtain medical attention.
Maximum accepted concentration	25 ppm (80 mg/m3)
TLV(ACGIH)	No data is available.
STEL	No data is available.
Toxicity by ingestion LD₅₀ (rat)	20 g/kg
Odour threshold	No data is available.
NFPA Hazard signals:	
Health	0
Flammability	1
Stability	----
Reactivity	0

4. FIRST AID MEASURES

Eyes:

Flush eyes with plenty of water for several minutes. Get medical attention if eye irritation persists.

Skin:

Wash skin with plenty of soap and water for several minutes. Get medical attention if skin irritation develops or persists.

Ingestion:

If irritation, headache, nausea, or drowsiness occurs, remove to fresh air. Get medical attention if breathing becomes difficult or respiratory irritation persists.

Other Instructions:

Aspiration of this product during induced emesis may result in severe lung injury. If evacuation of stomach is necessary, use method least likely to cause aspiration, such as gastric lavage after endotracheal intubation. Contact a Poison center for additional treatment information.

5. FIRE –FIGHTING MEASURES

Ignition Temperature – AIT(degrees C) : 371.1(700°F)

Flash Point (degrees C) : 100 (212°F) (CC)

Flammable Limits%

Lower Limit	:	2.6
Upper Limit	:	12.5

Recommended Fire Extinguishing Agents and Special Procedures:

Use water spray, dry chemical, foam, or carbon dioxide to extinguish flames. Use water spray to cool fire-exposed containers. Water or foam may cause frothing.

Unusual or Explosive Hazards: None

Special Protective Equipment for Fire Fighters: No special equipment or procedures required.

6. ACCIDENTAL RELEASE MEASURES (Transportation Spills):

Procedures in Case of Accidental Release, Breakage or Leakage:

Contain spill if possible, contain with absorbent materials such as clay or soil, and shovel up. Avoid skin and eye contact.

7. HANDLING AND STORAGE

Precautions to be taken in

Handling:

Keep the containers tightly closed when not in use.
Water contamination should be avoided.

Storage:

Store in a room temperature, well ventilated area in sealed containers.
Store away from strong oxidizing agent or combustible material.
Keep away from UV light as this may adversely affect the quality.

8. EXPOSURE CONTROLS/PERSONAL PROTECTION

Protective Equipment

Eye/Face/ Protection:

Safety glasses, Chemical type goggles, or face shield recommended to prevent eye contact.

Skin Protection:

Workers should wash exposed skin several times daily with soap and water. Soiled work clothing should be laundered or dry-cleaned.

Respiratory Protection:

Airborne concentrations should be kept to lowest levels possible. If vapor, mist or dust is generated and the occupational exposure limit of the product, or any component of the product, is exceeded, use appropriate NIOSH or MSHA approved air purifying or air supplied respirator after determining the airborne concentration of the contaminant. Air supplied respirators should always be worn when airborne concentration of the contaminant or oxygen content is unknown.

Ventilation:

Local exhaust ventilation recommended if generating vapor, dust, or mist. If exhaust ventilation is not available or inadequate, use MSHA or NIOSH approved respirator as appropriate.

9. PHYSICAL AND CHEMICAL PROPERTIES

Boiling Range / point	185 –189 ° C
Physical state	Liquid
Appearance	Clear colourless hygroscopic viscous liquid
Melting / freezing point	< -60° C
Vapour pressure @ 35°C (mm Hg)	0.07
Odour	Odourless
Vapour density (Air = 1)	2.62
Solubility in water @ 30°C	Miscible
Specific gravity (Water = 1)	1.036
pH	Not applicable
Others	----

10. STABILITY AND REACTIVITY

Chemical stability	Stable under normal storage conditions.
Incompatibility with other material	Acid chlorides, Acid anhydrides, oxidising agents, chloroformates, reducing agents & plastics
Reactivity with water	Not applicable
Hazardous Reaction products	None.

11. TOXICOLOGICAL INFORMATION

Toxicity by ingestion LD ₅₀ (rat)	20 g/Kg - Practically non-toxic
Odour threshold	No data is available.

12. DISPOSAL CONSIDERATIONS

Waste Disposal Methods:

This product has been evaluated for Resource Conservation and Recovery Act (RCRA) characteristics and does not meet the criteria of a hazardous waste if discarded in its purchased form. Under RCRA, it is the responsibility of the user of the product to determine at the time of disposal, whether the product meets RCRA criteria for hazardous waste. This is because product uses, transformations, mixtures, processes, etc. may render the resulting materials hazardous.

Remarks: None

13. TRANSPORT INFORMATION

Transportation

DOT:

Proper Shipping Name:

Not regulated

Hazard Class:

Not regulated

Identification Number:

Not regulated

Packing Group:

Not regulated

Label Required:

Not regulated

IMDG

Proper Shipping Name:

Not regulated

ICAO

Proper Shipping Name:

Not regulated

TDG

Proper Shipping Name:

Not regulated

Hazard Class:

Not regulated

Identification Number:

Not regulated

Label Required:

Not regulated

14. REGULATORY INFORMATION

DOT SHIPPING NAME:

Not designated as a hazardous material by the federal DOT

DOT Hazard Class: Not Applicable

DOT Identification Number : Not Applicable

SARA 311 Categories:

- | | |
|--------------------------------------|------|
| 1. Immediate (Acute) Health Effects | : No |
| 2. Delayed (Chronic) Health Effects | : No |
| 3. Fire Hazard | : No |
| 4. Sudden release of Pressure Hazard | : No |
| 5. Reactivity Hazard | : No |

Regulatory Lists Searched:

- 01. SARA 313
- 02. MASS RTK
- 03. WTP Carcinogen
- 04. MI 406
- 05. IARC Group 1
- 06. IARC Group 2A
- 07. IARC Group 2B
- 08. OSHA STEL
- 09. OSHATWA

None of the other components of the material are found on the regulating lists indicated above.

15. ENVIRONMENTAL INFORMATION

Aquatic toxicity:

LC50-96hr Aquatic toxicity rating is believed to be > 100.00mg/liter practically nontoxic.

Mobility:

Not determined.

Persistence and Biodegradability:

This product is reported to have a moderate($\geq 30\%$) rate of biodegradation in a test for ready biodegradation.

Potential to Bioaccumulate:

This product is reported to have a low potential to bioconcentrate.

16. OTHER INFORMATION

Disclaimer of liability

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